

CN Unit 3 — Network Layer & Routing Complete Notes

Computer Networks (CN) · Unit 3 · Network Layer & Routing · 2024

Overview

These notes cover **Network Layer & Routing**. From bits on the wire to HTTP — layered network architectures, routing, transport protocols and the applications on top. Each section states the core idea, a worked intuition and the points examiners look for.

1. IPv4 Addressing

IPv4 addresses are 32-bit, organised into classes or CIDR blocks.

Why it matters: IPv4 Addressing appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of ipv4 addressing in one line.
- A small example showing ipv4 addressing in practice.
- Common mistakes students make and how to avoid them.

2. Subnetting

Subnet masks split an address block into smaller networks for efficient allocation.

Why it matters: Subnetting appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of subnetting in one line.
- A small example showing subnetting in practice.
- Common mistakes students make and how to avoid them.

3. Distance Vector Routing

Routers share distance vectors with neighbours using the Bellman-Ford equation; count-to-infinity is a known issue.

Why it matters: Distance Vector Routing appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of distance vector routing in one line.
- A small example showing distance vector routing in practice.
- Common mistakes students make and how to avoid them.

4. Link State Routing

Each router floods link-state advertisements and runs Dijkstra's algorithm (OSPF).

Why it matters: Link State Routing appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of link state routing in one line.
- A small example showing link state routing in practice.
- Common mistakes students make and how to avoid them.

5. IPv6

IPv6 uses 128-bit addresses, a simplified header and extension headers.

Why it matters: IPv6 appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of ipv6 in one line.
- A small example showing ipv6 in practice.
- Common mistakes students make and how to avoid them.

6. ICMP, ARP and NAT

ICMP reports errors, ARP resolves IP to MAC addresses and NAT maps private to public addresses.

Why it matters: ICMP, ARP and NAT appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of icmp, arp and nat in one line.
- A small example showing icmp, arp and nat in practice.
- Common mistakes students make and how to avoid them.

Quick revision

IPv4 Addressing · Subnetting · Distance Vector Routing · Link State Routing · IPv6 · ICMP, ARP and NAT